

Norway

SLEIPNER

Volve F

F-5

F-5

Design: F-5

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table (As Installed) @ 54.90m
Site:	Volve F	MD Reference:	Rotary Table (As Installed) @ 54.90m
Well:	F-5	North Reference:	Grid
Wellbore:	F-5	Survey Calculation Method:	Minimum Curvature
Design:	F-5	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9		
Site Position:		Northing:	6,478,563.53 m
From:	Map	Easting:	435,050.02 m
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in
		Latitude:	58° 26' 29.8068 N
		Longitude:	1° 53' 14.9285 E
		Grid Convergence:	-0.95 °

Well	F-5		
Well Position	+N/-S	-4.65 m	Northing:
	+E/-W	0.92 m	Easting:
Position Uncertainty	0.00 m		Wellhead Depth:
			91.00 m
			Latitude:
			58° 26' 29.6569 N
			Longitude:
			1° 53' 14.9902 E
			Water Depth:
			91.00 m

Wellbore	F-5		
Magnetics	Model Name	Sample Date	Declination
			(°)
			Dip Angle
			(°)
			Field Strength
			(nT)
	MNETICREFERENCE	01.06.2008	-2.14
			71.62
			50,352

Design	F-5		
Audit Notes:			
Version:	1.0	Phase:	ACTUAL
		Tie On Depth:	145.90
Vertical Section:	Depth From (TVD)	+N/-S	+E/-W
	(m)	(m)	(m)
	145.90	-4.65	0.92
			Direction
			(°)
			86.76

Survey Program	Date	30.09.2017		
From	To	Survey (Wellbore)	Tool Name	Description
(m)	(m)			
146.00	2,775.50	F5 12¼" Drop Gyro (F-5)	Keeper, cont	Gyro Tool from SDC
2,775.50	2,913.31	F5 12 ¼" MWD Surveys (F-5)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)
2,927.67	3,761.77	F5 8 1/2" MWD surveys (F-5)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)

Survey										
Measured	Inclination	Azimuth	Vertical	+N/-S	+E/-W	Vertical	Dogleg	Build	Turn	
Depth	(°)	(°)	Depth	(m)	(m)	Section	Rate	Rate	Rate	
(m)			(m)			(m)	(°/30m)	(°/30m)	(°/30m)	
145.90	0.00	0.00	145.90	-4.65	0.92	0.00	0.000	0.000	0.000	
146.00	0.00	0.00	146.00	-4.65	0.92	0.00	0.000	0.000	0.000	
154.90	0.16	221.03	154.90	-4.66	0.92	-0.01	0.539	0.539	0.000	
195.30	0.28	98.15	195.30	-4.72	0.98	0.05	0.290	0.089	-91.248	
235.60	0.64	43.64	235.60	-4.57	1.23	0.31	0.394	0.268	-40.578	
276.00	1.51	188.03	275.99	-4.94	1.31	0.37	1.533	0.646	107.220	
316.40	2.21	178.31	316.37	-6.24	1.26	0.25	0.568	0.520	-7.218	
356.70	4.11	173.29	356.61	-8.45	1.45	0.31	1.428	1.414	-3.737	
397.10	7.66	171.91	396.79	-12.56	2.00	0.63	2.638	2.636	-1.025	

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Well:	F-5	North Reference:	Grid
Wellbore:	F-5	Survey Calculation Method:	Minimum Curvature
Design:	F-5	Database:	Production EDM P246N

Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
437.50	9.13	178.88	436.76	-18.43	2.44	0.74	1.326	1.092	5.176
477.80	10.58	180.10	476.46	-25.32	2.50	0.40	1.090	1.079	0.908
518.20	11.35	177.18	516.12	-33.00	2.69	0.16	0.705	0.572	-2.168
558.50	11.51	168.58	555.63	-40.91	3.68	0.70	1.273	0.119	-6.402
598.90	12.38	160.99	595.16	-48.95	5.89	2.45	1.332	0.646	-5.636
639.30	13.56	152.14	634.53	-57.24	9.51	5.60	1.713	0.876	-6.572
679.60	14.30	144.14	673.65	-65.45	14.64	10.25	1.534	0.551	-5.955
720.00	15.63	138.03	712.68	-73.54	21.20	16.35	1.531	0.988	-4.537
760.40	18.50	128.65	751.30	-81.59	29.85	24.53	2.947	2.131	-6.965
800.70	21.63	122.01	789.16	-89.52	41.14	35.36	2.878	2.330	-4.943
841.10	23.86	117.08	826.42	-97.19	54.73	48.49	2.177	1.656	-3.661
881.50	26.84	113.98	862.92	-104.62	70.35	63.66	2.422	2.213	-2.302
921.80	25.33	116.92	899.12	-112.22	86.35	79.21	1.479	-1.124	2.189
962.20	22.47	119.01	936.05	-119.88	100.81	93.21	2.215	-2.124	1.552
1,002.60	19.82	122.15	973.73	-127.27	113.36	105.33	2.139	-1.968	2.332
1,043.00	17.20	122.87	1,012.04	-134.16	124.18	115.74	1.953	-1.946	0.535
1,083.30	15.00	123.11	1,050.75	-140.24	133.55	124.75	1.638	-1.638	0.179
1,123.70	17.73	118.94	1,089.51	-146.08	143.32	134.17	2.206	2.027	-3.097
1,164.10	21.77	113.96	1,127.53	-152.10	155.55	146.05	3.247	3.000	-3.698
1,204.40	26.13	110.24	1,164.36	-158.21	170.72	160.84	3.433	3.246	-2.769
1,244.80	29.60	108.19	1,200.07	-164.40	188.55	178.30	2.673	2.577	-1.522
1,285.20	31.04	109.95	1,234.94	-171.07	207.82	197.16	1.256	1.069	1.307
1,325.60	28.60	112.14	1,269.99	-178.27	226.57	215.47	1.984	-1.812	1.626
1,365.90	25.50	114.90	1,305.88	-185.56	243.38	231.84	2.489	-2.308	2.055
1,406.30	29.89	114.65	1,341.64	-193.42	260.42	248.42	3.261	3.260	-0.186
1,446.60	31.10	113.78	1,376.37	-201.81	279.08	266.56	0.959	0.901	-0.648
1,487.00	31.91	113.02	1,410.81	-210.19	298.45	285.44	0.670	0.601	-0.564
1,527.30	31.93	113.02	1,445.02	-218.52	318.06	304.54	0.015	0.015	0.000
1,567.70	32.67	110.91	1,479.17	-226.59	338.08	324.08	1.001	0.550	-1.567
1,608.00	33.69	107.70	1,512.90	-233.87	358.89	344.44	1.512	0.759	-2.390
1,648.40	34.24	106.15	1,546.41	-240.44	380.49	365.63	0.762	0.408	-1.151
1,688.80	34.99	103.16	1,579.66	-246.24	402.68	387.46	1.379	0.557	-2.220
1,729.10	35.88	101.36	1,612.49	-251.20	425.52	409.98	1.021	0.663	-1.340
1,769.50	35.93	103.37	1,645.22	-256.27	448.66	432.79	0.876	0.037	1.493
1,809.80	35.95	104.77	1,677.85	-262.02	471.60	455.37	0.612	0.015	1.042
1,850.20	35.83	106.25	1,710.58	-268.35	494.42	477.80	0.650	-0.089	1.099
1,890.60	36.05	109.56	1,743.29	-275.64	516.97	499.91	1.452	0.163	2.458
1,930.90	36.80	111.45	1,775.72	-284.03	539.38	521.81	1.005	0.558	1.407
1,971.30	36.16	113.43	1,808.20	-293.19	561.58	543.45	0.995	-0.475	1.470
2,011.70	35.90	113.73	1,840.87	-302.70	583.36	564.66	0.233	-0.193	0.223
2,052.00	34.88	114.41	1,873.73	-312.22	604.67	585.40	0.814	-0.759	0.506
2,092.40	33.50	112.95	1,907.14	-321.34	625.46	605.64	1.192	-1.025	-1.084
2,132.70	32.25	110.97	1,940.99	-329.52	645.74	625.42	1.227	-0.931	-1.474

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Well:	F-5	North Reference:	Grid
Wellbore:	F-5	Survey Calculation Method:	Minimum Curvature
Design:	F-5	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
2,173.00	32.85	108.23	1,974.96	-336.79	666.16	645.40	1.185	0.447	-2.040	
2,213.40	32.96	103.95	2,008.89	-342.87	687.24	666.10	1.728	0.082	-3.178	
2,253.70	33.13	101.52	2,042.67	-347.71	708.67	687.23	0.994	0.127	-1.809	
2,294.10	33.06	98.75	2,076.52	-351.59	730.38	708.68	1.124	-0.052	-2.057	
2,334.40	33.27	95.46	2,110.25	-354.32	752.25	730.36	1.349	0.156	-2.449	
2,374.80	33.48	93.92	2,143.99	-356.13	774.40	752.37	0.648	0.156	-1.144	
2,415.20	33.45	93.04	2,177.69	-357.49	796.64	774.50	0.361	-0.022	-0.653	
2,455.60	33.33	91.93	2,211.43	-358.45	818.85	796.62	0.462	-0.089	-0.824	
2,496.00	32.61	92.01	2,245.32	-359.21	840.82	818.51	0.536	-0.535	0.059	
2,536.30	31.90	90.38	2,279.40	-359.66	862.32	839.96	0.836	-0.529	-1.213	
2,576.70	31.49	87.08	2,313.78	-359.19	883.54	861.16	1.323	-0.304	-2.450	
2,617.10	31.21	84.61	2,348.28	-357.67	904.50	882.17	0.977	-0.208	-1.834	
2,657.40	31.29	83.98	2,382.74	-355.59	925.30	903.06	0.250	0.060	-0.469	
2,697.80	31.42	85.86	2,417.24	-353.73	946.23	924.07	0.733	0.097	1.396	
2,738.20	31.03	84.46	2,451.78	-351.96	967.10	945.00	0.612	-0.290	-1.040	
2,775.50	30.68	81.14	2,483.81	-349.57	986.08	964.08	1.398	-0.282	-2.670	
2,789.43	30.65	80.78	2,495.79	-348.45	993.09	971.15	0.401	-0.065	-0.775	
2,829.57	30.89	78.69	2,530.28	-344.79	1,013.29	991.53	0.819	0.179	-1.562	
2,866.88	30.86	77.50	2,562.30	-340.84	1,032.03	1,010.45	0.492	-0.024	-0.957	
2,904.41	31.20	77.01	2,594.46	-336.58	1,050.90	1,029.54	0.339	0.272	-0.392	
2,913.31	31.86	76.70	2,602.05	-335.52	1,055.43	1,034.12	2.291	2.225	-1.045	
2,927.67	32.38	76.45	2,614.21	-333.75	1,062.86	1,041.63	1.121	1.086	-0.522	
2,954.56	34.03	71.62	2,636.71	-329.68	1,077.00	1,055.99	3.477	1.841	-5.389	
2,995.22	36.51	61.44	2,669.94	-320.30	1,098.44	1,077.92	4.702	1.830	-7.511	
3,036.42	36.77	55.01	2,703.01	-307.37	1,119.32	1,099.50	2.800	0.189	-4.682	
3,076.71	37.73	48.33	2,735.09	-292.25	1,138.41	1,119.42	3.093	0.715	-4.974	
3,117.15	37.11	43.61	2,767.21	-275.19	1,156.08	1,138.01	2.176	-0.460	-3.501	
3,157.49	35.73	34.79	2,799.70	-256.69	1,171.20	1,154.16	4.024	-1.026	-6.559	
3,197.04	34.20	28.91	2,832.12	-237.47	1,183.17	1,167.19	2.806	-1.161	-4.460	
3,236.87	31.59	21.36	2,865.57	-217.94	1,192.38	1,177.50	3.658	-1.966	-5.687	
3,253.79	30.18	19.04	2,880.09	-209.80	1,195.39	1,180.96	3.272	-2.500	-4.113	
3,278.34	28.32	16.57	2,901.51	-198.38	1,199.06	1,185.27	2.709	-2.273	-3.018	
3,318.98	30.91	22.37	2,936.85	-179.48	1,205.78	1,193.05	2.850	1.912	4.281	
3,359.21	34.92	27.52	2,970.62	-159.70	1,215.04	1,203.41	3.644	2.990	3.840	
3,398.47	40.44	30.90	3,001.68	-138.79	1,226.78	1,216.32	4.502	4.218	2.583	
3,439.55	44.73	31.42	3,031.92	-115.01	1,241.17	1,232.02	3.143	3.133	0.380	
3,479.32	47.93	28.18	3,059.38	-90.05	1,255.44	1,247.68	2.991	2.414	-2.444	
3,520.35	50.65	26.17	3,086.14	-62.38	1,269.63	1,263.42	2.279	1.989	-1.470	
3,559.87	53.98	23.14	3,110.30	-33.96	1,282.66	1,278.03	3.115	2.528	-2.300	
3,601.65	54.71	22.31	3,134.66	-2.65	1,295.77	1,292.89	0.714	0.524	-0.596	
3,642.07	54.84	22.15	3,157.97	27.92	1,308.26	1,307.09	0.137	0.096	-0.119	
3,682.43	55.13	22.24	3,181.13	58.52	1,320.75	1,321.29	0.222	0.216	0.067	
3,722.88	53.47	22.04	3,204.73	88.95	1,333.13	1,335.37	1.237	-1.231	-0.148	

Statoil

Survey Report

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Site:	Volve F	MD Reference:	Rotary Table (As Installed) @ 54.90m
Well:	F-5	North Reference:	Grid
Wellbore:	F-5	Survey Calculation Method:	Minimum Curvature
Design:	F-5	Database:	Production EDM P246N

Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
3,761.77	52.67	22.04	3,228.10	117.77	1,344.80	1,348.64	0.617	-0.617	0.000	
3,792.00	52.67	22.04	3,246.43	140.05	1,353.82	1,358.91	0.000	0.000	0.000	

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Well:	F-5	North Reference:	Grid
Wellbore:	F-5	Survey Calculation Method:	Minimum Curvature
Design:	F-5	Database:	Production EDM P246N

Design Targets									
Target Name - hit/miss target - Shape	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
F5 Tgt Rev 2 101106 - actual wellpath misses target center by 121.43m at 3321.99m MD (2939.43 TVD, -178.04 N, 1206.38 E) - Point	0.00	0.00	3,004.90	-263.61	1,150.38	6,478,300.00	436,200.00	58° 26' 21.8979 N	1° 54' 26.0898 E
F5 Pilot - actual wellpath misses target center by 1003.18m at 3241.06m MD (2869.15 TVD, -215.91 N, 1193.17 E) - Point	0.00	0.00	3,064.90	-843.82	1,950.66	6,477,720.00	437,000.00	58° 26' 3.5664 N	1° 55' 15.9815 E
F5 Top Hugin RTD - actual wellpath misses target center by 33.50m at 3560.07m MD (3110.42 TVD, -33.81 N, 1282.72 E) - Point	0.00	0.00	3,102.00	-40.54	1,314.44	6,478,523.00	436,364.00	58° 26' 29.1926 N	1° 54' 35.9771 E
F5 Hugin T1 230907 - actual wellpath misses target center by 81.47m at 3499.82m MD (3072.94 TVD, -76.43 N, 1262.58 E) - Point	0.00	0.00	3,104.55	-132.57	1,312.44	6,478,431.00	436,362.00	58° 26' 26.2176 N	1° 54' 35.9457 E
F5 Top Target 291007 - actual wellpath misses target center by 30.42m at 3597.99m MD (3132.54 TVD, -5.41 N, 1294.64 E) - Point	0.00	0.00	3,124.90	-11.53	1,323.44	6,478,552.00	436,373.00	58° 26' 30.1348 N	1° 54' 36.5030 E
Volve I30 Topp - actual wellpath misses target center by 182.82m at 3559.87m MD (3110.30 TVD, -33.96 N, 1282.66 E) - Point	0.00	0.93	3,133.83	-121.57	1,441.38	6,478,442.00	436,490.90	58° 26' 26.6407 N	1° 54' 43.8814 E
F5 040107 - actual wellpath misses target center by 54.38m at 3671.14m MD (3174.67 TVD, 49.95 N, 1317.25 E) - Point	0.00	0.00	3,134.90	66.50	1,350.45	6,478,630.00	436,400.00	58° 26' 32.6704 N	1° 54' 38.0896 E
F5 1809 Pt 3 - actual wellpath misses target center by 202.51m at 3507.47m MD (3077.90 TVD, -71.24 N, 1265.22 E) - Point	0.00	0.00	3,192.70	-220.50	1,339.75	6,478,343.10	436,389.30	58° 26' 23.3905 N	1° 54' 37.7166 E
F5 1809 Pt 2 - actual wellpath misses target center by 156.21m at 3600.08m MD (3133.75 TVD, -3.83 N, 1295.29 E) - Point	0.00	0.00	3,217.80	-104.66	1,379.96	6,478,458.90	436,429.50	58° 26' 27.1549 N	1° 54' 40.0792 E
Volve I31 Bunn - actual wellpath misses target center by 214.61m at 3617.11m MD (3143.58 TVD, 9.03 N, 1300.56 E) - Point	0.00	0.93	3,239.34	-121.57	1,441.38	6,478,442.00	436,490.90	58° 26' 26.6407 N	1° 54' 43.8814 E
F5 1809 Pt 1 - actual wellpath misses target center by 81.56m at 3725.21m MD (3206.12 TVD, 90.68 N, 1333.83 E) - Point	0.00	0.00	3,240.10	40.89	1,388.76	6,478,604.40	436,438.30	58° 26' 31.8629 N	1° 54' 40.4765 E
F5 Base Hugin RTD - actual wellpath misses target center by 17.19m at 3791.25m MD (3245.98 TVD, 139.50 N, 1353.59 E) - Rectangle (sides W47.00 H65.00 D0.00)	25.00	65.00	3,243.00	135.31	1,370.00	6,478,698.79	436,419.54	58° 26' 34.9040 N	1° 54' 39.2256 E
F5 Target Base 291007 - actual wellpath misses target center by 17.08m at 3792.00m MD (3246.43 TVD, 140.05 N, 1353.82 E) - Point	0.00	0.00	3,244.90	136.52	1,370.46	6,478,700.00	436,420.00	58° 26' 34.9437 N	1° 54' 39.2528 E
F5 Bae Hugin T2 230907 - actual wellpath misses target center by 17.43m at 3792.00m MD (3246.43 TVD, 140.05 N, 1353.82 E) - Point	0.00	0.00	3,251.00	142.53	1,370.46	6,478,706.00	436,420.00	58° 26' 35.1376 N	1° 54' 39.2468 E

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Project:	SLEIPNER	TVD Reference:	Rotary Table (As Installed) @ 54.90m
Site:	Volve F	MD Reference:	Rotary Table (As Installed) @ 54.90m
Well:	F-5	North Reference:	Grid
Wellbore:	F-5	Survey Calculation Method:	Minimum Curvature
Design:	F-5	Database:	Production EDM P246N

Casing Points					
Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)	
220.80	220.80	30"	30.000	36.000	
1,404.70	1,340.25	13 3/8"	13.375	17.500	
2,921.00	2,608.57	9 5/8"	9.625	12.250	
3,790.10	3,245.28	7"	7.000	8.500	

Formations					
Measured Depth (m)	Vertical Depth (m)	Name	Lithology	Dip (°)	Dip Direction (°)
898.35	878.00	Utsira Top	Empty	0.00	
1,098.08	1,065.00	Utsira Base	Empty	0.00	
1,273.62	1,225.00	Skade Top	Empty	0.00	
1,377.17	1,316.00	Skade Base	Empty	0.00	
2,301.84	2,083.00	Grid Top	Empty	0.00	
2,382.00	2,150.00	Grid Base	Empty	0.00	
2,521.67	2,267.00	Balder Top	Empty	0.00	
2,602.73	2,336.00	Sele Top	Empty	0.00	
2,677.61	2,400.00	Lista Top	Empty	0.00	
2,783.86	2,491.00	Ty Top	Empty	0.00	
2,899.19	2,590.00	Ekofisk Top	Empty	0.00	
2,928.61	2,615.00	Tor Top	Empty	0.00	
3,084.18	2,741.00	Hod Top	Empty	0.00	
3,179.91	2,818.00	Blodoeks Top	Empty	0.00	
3,235.02	2,864.00	Hidra Top	Empty	0.00	
3,247.88	2,875.00	Roedby Top	Empty	0.00	
3,257.15	2,883.00	Aasgard Top	Empty	0.00	
3,306.39	2,926.00	Draupne Top	Empty	0.00	
3,400.20	3,003.00	Heather Top	Empty	0.00	
3,472.81	3,055.00	Hugin Top	Empty	0.00	
3,713.20	3,199.00	Sleipner Top	Empty	0.00	

Design Annotations				
Measured Depth (m)	Vertical Depth (m)	Local Coordinates		Comment
		+N/-S (m)	+E/-W (m)	
3,792.00	3,246.43	140.05	1,353.82	Projection to TD

Checked By: _____ Approved By: _____ Date: _____